



CNET123

FINAL PROJECT

GROUP 3A

Dannang, Gabriel
Novelo, Vic
Elorde, John Aaron
Parikh, Vandit
Khan, Muhammad

Contents

Introduction	2
Background	2
Project Scope.....	2
Assumptions	2
Client Requirements Analysis	2
Department and Subnet Requirements	2
Printer Allocation Per Department	3
Security, Scalability, and Durability Needs.....	3
CISCO Network Design.....	4
Network Topology	4
IP Addressing and Subnetting Plan	6
Network Devices and Cost Estimation.....	11
AWS VPC Design	16
VPC Architecture	16
Cost Estimation Over 10 Years	18
Comparative Analysis	20
Cost Analysis.....	20
Scalability & Flexibility	21
Security & Compliance	22
Performance & Reliability	22
Final Recommendation: Hybrid Network Solution	23
Justification for the Recommended Approach.....	24
Implementation Roadmap	25
Conclusion	26
Final Decision:	26
References	27

Introduction

Background

A mid-sized FMCG company in Regina, Saskatchewan, distributes packaged foods and household products across the province. From its headquarters, it has nine departments—Marketing, Sales, HR, Technical, Customer Service, Financial, Strategy, Legal, and Quality Control—totaling 1,258 employees, with up to 20% growth expected in key areas like Customer Service. The company requires a secure, scalable network for efficient communication, shared printers (one per 30 employees in larger departments, or one per department for smaller ones), and 10 Gb performance to support operations, sales growth, and customer service excellence.

Project Scope

Our consulting group will design two network solutions for the FMCG company: one on-premise network using Cisco equipment, and one cloud network using Amazon Web Services (AWS) Virtual Private Cloud (VPC). This report will fully outline both solutions, including IP addressing, subnetting, a logical star topology for increased security and scalability, configuration of AWS along with supporting documentation, and a cost analysis over a ten-year period. We will compare the solutions against the following metrics: cost, scalability, maintenance, security, and performance, before making a recommendation for the most appropriate strategy—be it Cisco, AWS, or a hybrid solution—to meet the company's corporate goals. In addition, a group presentation will briefly present the findings to the company's leadership team.

Assumptions

We assume the FMCG company operates from a single headquarters in Regina, with all departments co-located to streamline operations and communication. The company's growth projections require a scalable network design. Existing Category 6A cabling supports 10 Gb Ethernet across departments, eliminating cabling costs. The client can secure a public IPv4 address, and we recommend a Class B (e.g., 172.16.0.0/16) for its 65,534-host capacity, fitting 1,258 employees and growth, as Class C (254 hosts) is too small for Customer Service (960+ hosts). We also assume standard security needs without specific compliance requirements, focusing on network-level protections.

Client Requirements Analysis

Department and Subnet Requirements

The client is a Fast-Moving Consumer Goods (FMCG) organization with the following departments:

Department	Employees	Expected Growth	Total (w/ Growth)	Subnet Needed
Marketing	40	-	40	Yes

Sales	200	10%	220	Yes
HR	50	-	50	Yes
Technical	30	-	30	Yes
Customer Service	800	20%	960	Yes
Financial	20	-	20	Yes
Strategy	10	-	10	Yes
Legal	8	-	8	Yes
Quality Control	100	10%	110	Yes

We will create 9 separate subnets, one for each department.

Printer Allocation Per Department

Printers are allocated based on the rule 1 printer per 30 employees.

Department	Total Employees	Printers Needed
Marketing	40	2
Sales	220	8
HR	50	2
Technical	30	1
Customer Service	960	32
Financial	20	1
Strategy	10	1
Legal	8	1
Quality Control	110	4

We will need 52 Printers.

Security, Scalability, and Durability Needs

The client has emphasized:

- High Security: Each department must be isolated to prevent unauthorized access.
- Scalability: Ability to support growing departments
- Durability: The network must run continuously with minimal downtime.

To meet these requirements:

- VLANs or Subnets for segmentation

- Firewalls / ACLs for protection
- Redundant links/devices to avoid single points of failure

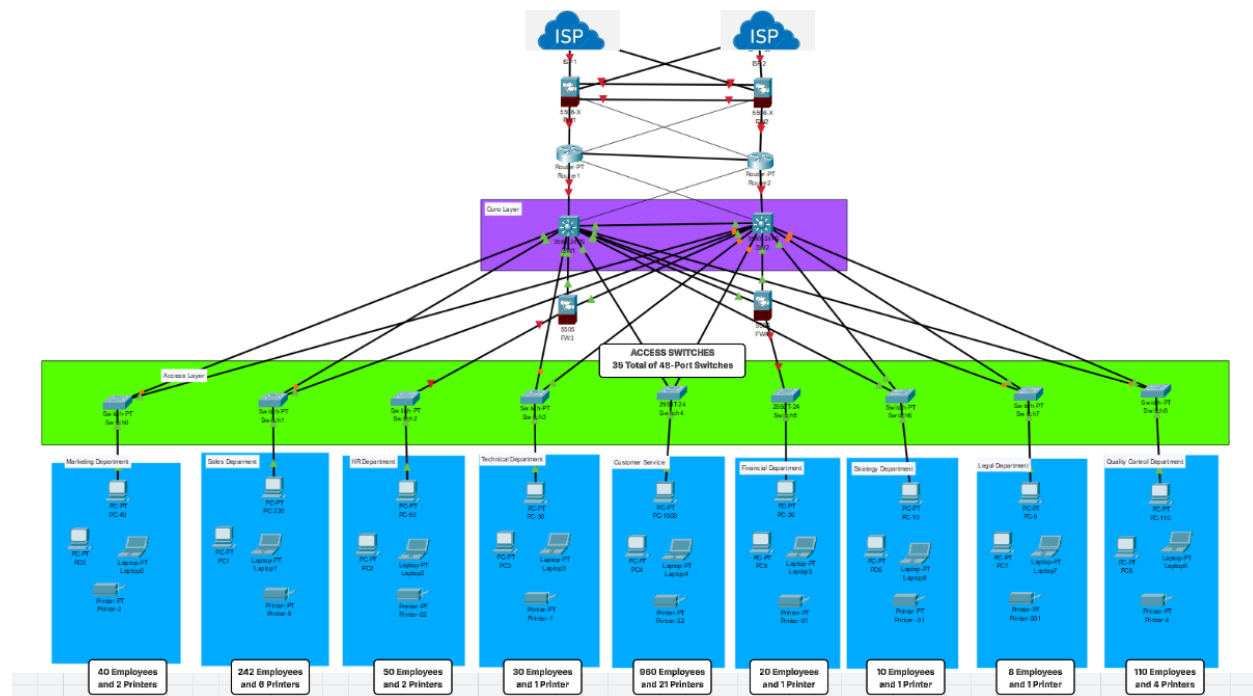
CISCO Network Design

Network Topology

Design Overview and Diagram

The proposed network design adopts a hybrid topology—implementing a **star topology at the Access Layer** and a **partial mesh at the Core Layer**. This two-tier physical structure aligns with organizational priorities of performance, reliability, security, and future scalability, while minimizing unnecessary complexity and maintenance costs.

The logical structure incorporates an additional perimeter layer, forming a three-tier logical model, with secure external connectivity managed separately from internal data operations.



Topology Structure

Access Layer - Star Topology

The access layer follows a classic **star topology**, where each department is connected to a dedicated 48-port access switch. A total of **35 switches** are used, ensuring that no switch is overloaded and that each department receives dedicated bandwidth and localized fault domains.

- Departments with higher device counts, such as **Customer Service, Sales, and Quality Control**, are allocated multiple switches to accommodate current and future user growth.

- Smaller departments like **Strategy, Legal, Finance**, and **Technical** are provided with individual switches as well to maintain simplicity, security, and predictable performance.
- **No access switch is shared** across departments, avoiding cross-traffic, reducing complexity, and improving administrative clarity.
- Each access switch includes **uplinks to both core switches**, enabling link redundancy and supporting load balancing.

This configuration ensures:

- Ease of maintenance and troubleshooting, as each switch serves a defined group of users
- High reliability, as local switch issues won't impact other departments
- Scalability, as new switches can be added per department without reengineering the network

Core Layer - Partial Mesh Topology

At the center of the design are two core switches, operating in a **partial mesh topology**. These devices provide high-speed interconnectivity between departments and act as the backbone of the entire network.

- **Redundant links** between the core switches and all access switches prevent single points of failure and ensure **continuous network availability**
- All **inter-VLAN routing** is performed at the core, separating switching functions from routing tasks for better performance and more efficient traffic management
- The **partial mesh structure** provides optimal paths for traffic between departments, reducing latency and bottlenecks, especially for high-throughput applications

Benefits of this design:

- Improved network efficiency, as traffic is centrally routed and optimized
- Simplified policy implementation, especially with VLAN-based access control
- Future-ready architecture, allowing seamless expansion as departmental demands grow

Perimeter Layer (Logical)

While the network is physically a two-tier structure (Access and Core), a **logical third layer**—the **Perimeter Layer**—is implemented to handle secure external communications.

This includes:

- **Two enterprise-grade edge routers** for WAN and Internet connectivity
- **Four firewalls** placed strategically:
 - **Two at the edge** (border) between external connections and the internal network
 - **Two internally**, between core components and sensitive departmental VLANs or services (e.g., servers, finance, HR)

The firewalls operate in **active-passive pairs**, ensuring high availability and seamless failover. Their role is to:

- Enforce **access control policies**
- Provide **intrusion prevention and detection**
- **Segment critical systems** to reduce lateral movement in case of breach

This logical tier adds a robust layer of **defense-in-depth**, aligning with best practices for enterprise-grade security and risk management.

Topology Summary

Physical Topology

- **Architecture:** Two-tier physical (Access + Core)
- **Design Patterns:**
 - **Access Layer:** Star topology per department
 - **Core Layer:** Partial mesh topology with redundancy

Advantages:

- Localizes issues at the access layer
- Enhances core layer resilience through redundancy
- Eases network expansion and maintenance

Logical Topology

- **Architecture:** Three-tier logical (Access, Core, Perimeter)
- **Key Features:**
 - Department-based **VLAN segmentation**
 - **Inter-VLAN routing** at the core
 - **Firewall filtering** at both perimeter and internal layers
 - Core switches act as the central routing and switching hub

Advantages:

- Improves internal traffic segmentation and security
- Facilitates policy enforcement per department
- Reduces network-wide broadcast traffic

IP Addressing and Subnetting Plan

IPv4 Address Selection and Class Justification

Objective

The primary goal of this subnetting plan is to allocate IP addresses efficiently across nine departments with varying sizes, ensuring scalability for future growth, robust security through network segmentation, and optimal utilization of the IPv4 address space.

Analysis of IPv4 Classes

After evaluating the three private IPv4 address classes, we determined that **Class B (172.16.0.0/16)** is the most suitable for this organization. While Class A (10.0.0.0/8) offers excessive addresses (16 million hosts) and would lead to significant waste, Class C (192.168.0.0/24) is too restrictive, supporting only 254 hosts per subnet—insufficient for the largest department (Customer Service), which requires 960+ hosts. Class B strikes the perfect balance, providing 65,534 hosts total while allowing flexible subnetting through VLSM (Variable Length Subnet Masking).

Decision

- **Selected Range:** 172.16.0.0/16 (Class B).
- **Justification:**
 - **Scalability:** Accommodates the largest department (Customer Service) and future 20% growth.
 - **Efficiency:** VLSM enables tailored subnet sizes, preventing IP waste.
 - **Security:** Isolates departments into separate subnets to limit broadcast domains and reduce attack surfaces.

Subnetting Calculations (number of subnets, hosts per subnet).

Methodology

We employed **Variable Length Subnet Masking (VLSM)** to customize subnet masks per department. This approach involves:

1. Calculating Host Requirements

- Round up each department's employee count (+ projected growth) to the nearest power of 2.
- *Example:*
 - Marketing: 40 employees → 64 hosts (2^6) → /26 subnet.
 - Customer Service: 960 employees → 1024 hosts (2^{10}) → /22 subnet.

2. Assigning Subnet Masks

- Larger subnets (e.g., /22) for high-growth departments.
- Smaller subnets (e.g., /28) for departments with few employees.

Why VLSM?

We rejected fixed-length subnetting because:

1. IP Address Waste

- Fixed masks would force small departments (e.g., Legal: 8 hosts) into oversized subnets (e.g., /22 wastes 1008 IPs).

2. Scalability Limitations

- High-growth departments (Customer Service: 960+ hosts) require subnetting that allows easy expansion.

3. Inefficient Resource Use

- A Class B network (65,534 hosts) would be exhausted prematurely if all subnets are uniformly large.

Benefits of VLSM:

- **Efficiency:** No wasted IPs (e.g., Legal uses only 16 IPs via /28).
- **Flexibility:** Subnets are tailored to departmental needs.
- **Future-Proofing:** Reserved address spaces for potential expansion.

Subnet Allocation Table

Based on the above methodology, the following table summarizes the customized subnet allocations for each department:

Department	Employees (Current)	Employees (Future)	Hosts Required	Subnet Mask	Usable Hosts	Network ID
Marketing	40	40	64	/26 (255.255.255.192)	62	172.16.0.0/26
Sales	200	220	256	/24 (255.255.255.0)	254	172.16.1.0/24
HR	50	50	64	/26 (255.255.255.192)	62	172.16.2.0/26
Technical	30	30	32	/27 (255.255.255.224)	30	172.16.3.0/27
Customer Service	800	960	1024	/22 (255.255.252.0)	1022	172.16.4.0/22
Financial	20	20	32	/27 (255.255.255.224)	30	172.16.8.0/27
Strategy	10	10	16	/28 (255.255.255.240)	14	172.16.9.0/28
Legal	8	8	16	/28 (255.255.255.240)	14	172.16.9.16/28
Quality Control	100	110	128	/25 (255.255.255.128)	126	172.16.10.0/25

Address Ranges and Critical IDs for Selected Subnets

To demonstrate the subnetting scheme, we examine the **first**, **second**, and **last** subnets from our allocation:

(1) First Subnet: Marketing (172.16.0.0/26)

Field	Value	Notes
Network ID	172.16.0.0	First address in the subnet.
First Usable Host	172.16.0.1	First assignable device address.
Last Usable Host	172.16.0.62	Supports 62 devices (Marketing: 40 employees).
Broadcast ID	172.16.0.63	Reserved for broadcast traffic.

(2) Second Subnet: Sales (172.16.1.0/24)

Field	Value	Notes
Network ID	172.16.1.0	Base address for Sales.
First Usable Host	172.16.1.1	Start of assignable range.
Last Usable Host	172.16.1.254	Supports 254 devices (Sales: 220 employees + growth).
Broadcast ID	172.16.1.255	Broadcast address for this subnet.

(3) Last Subnet: Legal (172.16.9.16/28)

Field	Value	Notes
Network ID	172.16.9.16	Starting address for Legal.
First Usable Host	172.16.9.17	First valid host IP.
Last Usable Host	172.16.9.30	Supports 14 devices (Legal: 8 employees + reserves).
Broadcast ID	172.16.9.31	Subnet broadcast address.

Key Observations and Technical Justifications

1. Progressive Subnet Sizes

Subnet Size	Department Example	Technical Rationale	Business Benefit
/26 (62 hosts)	Marketing	40 employees need ≤ 64 addresses (2^6). Next size (/25) would waste 50% of addresses.	Prevents IP exhaustion while maintaining room for 50% growth.
/24 (254 hosts)	Sales	220 employees + 10% growth = 242 needed. /24 is the smallest fit (vs. /23 would waste 510 IPs).	Optimizes address usage for medium-large departments.

/28 (14 hosts)	Legal	Only 8 devices needed. Next size (/27) would waste 60% of addresses.	Maximizes address space efficiency for micro-departments.
-----------------------	-------	--	---

Why This Matters:

VLSM enables *right-sized* allocations - no department gets more IPs than needed, preserving addresses for future expansions or new departments.

2. Non-Overlapping Ranges

- **Implementation Check:**

- Marketing: 172.16.0.0/26 → Ends at 172.16.0.63
- Sales: 172.16.1.0/24 → Starts at 172.16.1.0
- Legal: 172.16.9.16/28 → Fits between 172.16.9.16-172.16.9.31

- **Collision Avoidance:** Used sequential Network IDs with buffer space between subnets to prevent accidental overlaps during future expansions.

Practical Impact:

Eliminates network conflicts that could cause routing errors or downtime.

3. Broadcast Domain Isolation

- **Technical Behavior:**

- Broadcast traffic (e.g., ARP requests) stops at each subnet's Broadcast ID (e.g., 172.16.0.63 for Marketing).
- Devices in Sales (172.16.1.0/24) cannot receive Marketing's broadcast traffic.

- **Security Advantage:**

Limits the "blast radius" of broadcast storms and contains potential malware propagation.

4. Reserved Expansion Space

- **Strategic Gaps:**

Left unused address ranges between subnets (e.g., 172.16.0.64-172.16.0.255 after Marketing) for:

- Future department growth
- New network segments (Wi-Fi, IoT)
- Temporary project teams

Example:

If Marketing grows beyond 62 devices, we can assign 172.16.0.64/26 without renumbering.

5. Compliance with RFC 1918

All addresses use the **private 172.16.0.0/12** range (per RFC 1918), ensuring:

- No conflicts with public Internet addresses
- Safe NAT implementation at the network edge
- Standard compliance for enterprise networks

Security and Routing Strategy

These subnets will be routed through the core switch, with access policies enforced via the perimeter firewalls to ensure departmental segmentation and external traffic control. This setup strengthens security boundaries between departments and enables centralized management of inbound and outbound network traffic.

Network Devices and Cost Estimation

In this section, we outline the critical network devices that will form the backbone of the company's infrastructure, ensuring high availability, security, and scalability. The following devices—routers, core switches, access switches, and firewalls—have been selected based on their performance, reliability, and future-proof capabilities to meet the needs of both current and expanding departmental requirements.

Network Device Requirements

Routers

The router infrastructure ensures robust connectivity between internal networks and external resources, while supporting redundancy and failover capabilities for high availability.

Cisco Catalyst 8300 Series (Dual Router Configuration)

- **Role:** Core and Edge Routers
- **Key Features:** SD-WAN ready, high availability, multi-ISP failover, modular WAN/LAN interfaces, 10G uplinks.
- **Units Required:** 2
- **Cost:** \$6,885.00 each (Amazon Canada)
- **Justification:** The Cisco Catalyst 8300 routers deliver enterprise-grade performance with redundancy and failover capabilities, ensuring seamless connectivity for critical services like VoIP and cloud applications. They support modular, high-speed interfaces, which allow for easy future scalability. The dual-router configuration provides load balancing, optimizes network traffic, and eliminates single points of failure, making the network highly resilient.
- **Where Deployed:** These routers will be deployed at the **core** of the network to connect all internal department networks to external resources and ensure redundancy for high availability.

Core Switches

Core switches handle traffic aggregation between access switches and routers, forming the backbone of the network. These switches manage high-volume data traffic and ensure efficient routing to all departments.

Cisco Catalyst 9300-48P

- **Role:** Core Switch
- **Key Features:** 48-port PoE+ support, 10G uplink, advanced security features, energy-efficient, scalable architecture.
- **Units Required:** 2
- **Cost:** \$3,784.00 each (Amazon Canada)
- **Justification:** The Cisco Catalyst 9300-48P is ideal for core switching as it offers high-density PoE+ support and 10G uplinks, which are essential for high-speed, high-traffic aggregation. Its energy efficiency and advanced security features ensure the integrity and performance of the core network. These switches are built to scale, allowing easy integration of future network expansions.
- **Where Deployed:** These core switches will be deployed at the **central aggregation layer** of the network, interconnecting all access switches and routers, ensuring efficient data flow and high-speed connectivity.

Access Switches

Access switches provide the connectivity layer between user devices and the core network, offering Power over Ethernet (PoE) for devices such as phones, cameras, and wireless access points. These switches are distributed across departments based on size.

Department	Employees	Ports Needed	48-Port Switches
Marketing	40	40 + 2 printers = 42	1
Sales	220 + 10% = 242	242 + 9 printers = 251	6
HR	50	50 + 2 printers = 52	1
Technical	30	30 + 1 printer = 31	1
Customer Service	800 + 20% = 960	960 + 32 printers = 992	21
Financial	20	20 + 1 printer = 21	1
Strategy	10	10 + 1 printer = 11	1
Legal	8	8 + 1 printer = 9	1
Quality Control	100 + 10% = 110	110 + 4 printers = 114	3
Total	1,470	1,523	35

Cisco Catalyst 9200-48P

- **Role:** Access Switch
- **Key Features:** 48-port PoE+ support, 10G uplink, advanced security features, energy-efficient.
- **Units Required:** 35
- **Cost:** \$3,035.05 each (Amazon Canada)
- **Justification:** The Cisco Catalyst 9200-48P switches are deployed in department-specific locations to provide high-density port availability for end-user devices, printers, and other critical equipment. Their PoE+ capability reduces the need for additional power adapters, streamlining deployment and reducing operational costs. The 10G uplinks ensure efficient communication between departments, while security features protect the network from internal threats.
- **Where Deployed:** These switches will be deployed in **departmental network closets**, directly supporting the user devices in each department such as Marketing, Sales, HR, Technical, Customer Service, etc.

Firewalls

Firewalls secure the network by controlling incoming and outgoing traffic based on predefined security rules, protecting against external threats and enforcing internal network policies.

Cisco Firepower 1120

- **Role:** Next-Generation Firewall (NGFW)
- **Key Features:** Intrusion prevention, VPN support, content filtering, advanced threat defense, scalability.
- **Units Required:** 4
- **Cost:** \$1,950.00 each (Amazon Canada)
- **Justification:** The Cisco Firepower 1120 firewalls provide robust protection against external and internal threats, with advanced features like intrusion prevention, content filtering, and secure VPN support. The firewalls enable segmentation between departments and safeguard sensitive data, ensuring the security of the entire network. Their scalability allows for future expansion as the network grows, and they offer comprehensive visibility and control over network traffic.
- **Where Deployed:** The firewalls will be deployed at the **network perimeter** to manage inbound and outbound traffic and to segment internal departments for enhanced security. One firewall will be deployed per **departmental segment** to ensure optimal security and traffic filtering.

Cost Breakdown

Initial Hardware Costs

Device	Quantity	Unit Cost (Amazon)	Total Cost
Catalyst 9200-48P	35	\$3,035.05	\$106,226.75
Catalyst 9300-48P (Core)	2	\$3,784.00	\$7,568.00
Catalyst 8300 Router	2	\$6,885.00	\$13,770.00
Firepower 1120 Firewall	4	\$1,950.00	\$7,800.00
Total Hardware Cost			\$135,364.75

Replacement Costs

Anticipated replacement of devices based on standard industry lifecycles:

Device	Quantity	Replacement Window	Replacement Cost
Catalyst 9200-48P (Access)	35 (50%)	Year 6 and Year 9	\$106,226.75
Catalyst 9300-48P (Core)	2	Year 5	\$7,568.00
Catalyst 8300 Router	2	Year 5	\$13,770.00
Firepower 1120 Firewall	4 (50%)	Year 5 and Year 10	\$7,800.00
Total Replacement Costs			\$135,364.75

Additional Costs

Setup & Labor Cost:

- Network Engineer Rate: \$60/hour
- Estimated Setup Time: 80 hours
- Total Labor Cost: **\$4,800.00**

Maintenance Costs (10 Years):

- Annual Maintenance Cost (5% of hardware): \$6,768.24/year
- 10-Year Maintenance Cost: **\$ 67,682.40**

Final Cost Summary

Category	Cost
Initial Hardware	\$135,364.75
Replacement Costs	\$135,364.75
Setup Labor	\$4,800.00
10-Year Maintenance	\$67,682.40
Total Cost	\$343,211.90

Cost Assumptions

1. Device Lifecycles:

- Access Switches (Catalyst 9200-48P): Assumed lifecycle of 6-8 years. Replacement planned in two phases: 17 switches in Year 6 and 18 switches in Year 9.
- Core Switches (Catalyst 9300-48P): Assumed lifecycle of 5-7 years. Full replacement planned for Year 5 (both units).
- Routers (Catalyst 8300): Assumed lifecycle of 5-7 years, with full replacement planned for Year 5.
- Firewalls (Firepower 1120): Assumed lifecycle of 5-7 years. 50% replacement planned for Years 5 and 10.

2. Replacement Quantities:

- Catalyst 9300-48P (Core Switches): Full replacement in Year 5 (2 units).
- Catalyst 8300 Routers: Full replacement in Year 5 (2 units).
- Firepower 1120 Firewalls: 50% replacement of firewalls planned in Year 5 and 10 (2 units).

3. Hardware Pricing:

- Prices sourced from Amazon and assumed to remain constant over the 10-year period. Adjustments for inflation or discounts on newer models are not included.

4. Maintenance Costs:

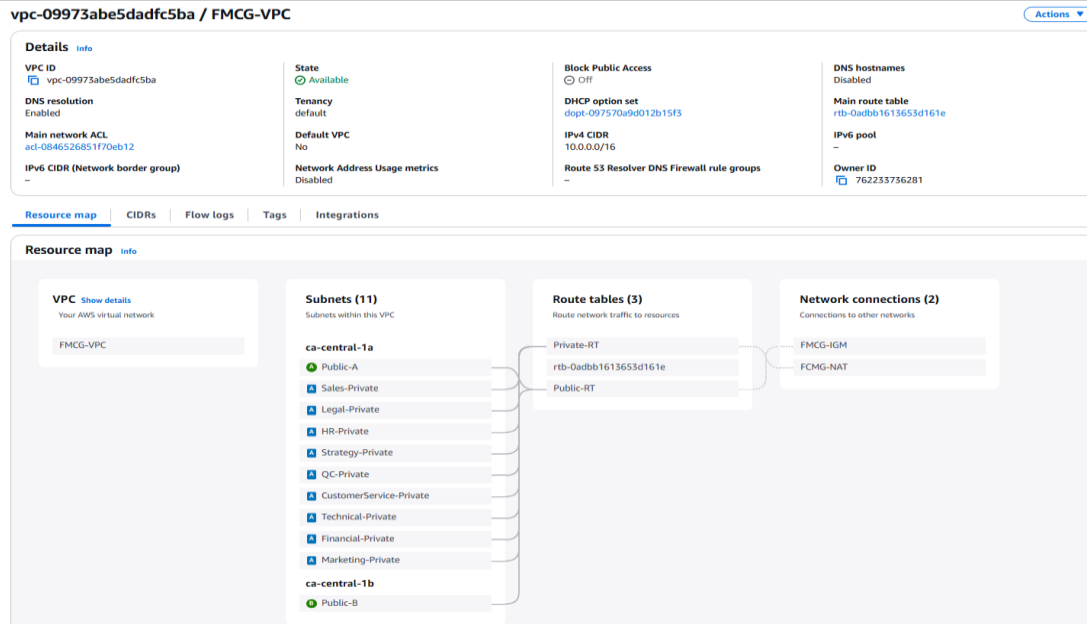
- Annual maintenance is estimated at 5% of initial hardware costs, not factoring in aging devices or increased costs near EOL.

5. Labor Costs:

- The network engineer labor rate of \$60/hour is based on average rates in Regina, Saskatchewan, and assumes 80 hours for the complete network installation.

AWS VPC Design

VPC Architecture



What it Shows:

- Resource map
- Attached subnets and destinations

How This Was Done:

1. Created Internet Gateway (IGW):

- VPC > Internet Gateways > Create
- Attached to FMCG-VPC

2. Created NAT Gateway:

- In a **Public Subnet**
- Assigned an **Elastic IP (EIP)**
- Created under **VPC > NAT Gateways**

3. Public Route Table:

- Destination: **0.0.0.0/0** → Target: **Internet Gateway**
- Associated with subnets: Public-A, Public-B

4. Private Route Table:

- Destination: **0.0.0.0/0** → Target: **NAT Gateway**
- Associated with subnets like: Sales-Private, Legal-Private, etc.

Why This Matters:

Public subnets can send/receive traffic to/from the internet. **Private subnets** can **only send traffic out**, via the **NAT Gateway**, and cannot receive inbound traffic. This setup ensures **security + functionality**, giving private servers internet access without exposing them directly.

Subnets (14) [Info](#)

Last updated 6 minutes ago [Actions](#) [Create subnet](#)

Find resources by attribute or tag

☐	Name	Subnet ID	State	VPC	Block Public...	IPv4 CIDR	IPv6 ...	IPv6 ...	Avail...	Availability Zone
☐	-	subnet-0aebac19f99adabc9	Available	vpc-0a060fb8779e5b232	Off	172.31.32.0/20	-	-	4091	ca-central-1d
☐	-	subnet-0d811c494173862bc	Available	vpc-0a060fb8779e5b232	Off	172.31.0.0/20	-	-	4091	ca-central-1b
☐	-	subnet-0e7c4bb857628ecf	Available	vpc-0a060fb8779e5b232	Off	172.31.16.0/20	-	-	4091	ca-central-1a
☐	Marketing-Private	subnet-03819e190dd567780	Available	vpc-09973abe5dadfc5ba FMC...	Off	10.0.1.0/26	-	-	59	ca-central-1a
☐	Sales-Private	subnet-074ce6661d274eb79	Available	vpc-09973abe5dadfc5ba FMC...	Off	10.0.2.0/24	-	-	251	ca-central-1a
☐	HR-Private	subnet-0574c0099b4ae42c0	Available	vpc-09973abe5dadfc5ba FMC...	Off	10.0.3.0/26	-	-	59	ca-central-1a
☐	Technical-Private	subnet-08aca7eb429a67645	Available	vpc-09973abe5dadfc5ba FMC...	Off	10.0.4.0/27	-	-	27	ca-central-1a
☐	Financial-Private	subnet-0a726d5dce6e2709c	Available	vpc-09973abe5dadfc5ba FMC...	Off	10.0.6.0/27	-	-	27	ca-central-1a
☐	Strategy-Private	subnet-0c702dbbb2e53714b	Available	vpc-09973abe5dadfc5ba FMC...	Off	10.0.7.0/28	-	-	11	ca-central-1a
☐	Legal-Private	subnet-0132118c1a42bfc51	Available	vpc-09973abe5dadfc5ba FMC...	Off	10.0.7.16/28	-	-	11	ca-central-1a
☐	QC-Private	subnet-02129e76c45868358	Available	vpc-09973abe5dadfc5ba FMC...	Off	10.0.12.0/25	-	-	123	ca-central-1a
☐	CustomerService-Private	subnet-050002d06a4c6d468	Available	vpc-09973abe5dadfc5ba FMC...	Off	10.0.8.0/22	-	-	1019	ca-central-1a
☐	Public-A	subnet-0f365c9d69e38c4f6	Available	vpc-09973abe5dadfc5ba FMC...	Off	10.0.100.0/24	-	-	250	ca-central-1a
☐	Public-B	subnet-0095820495ec3fe91	Available	vpc-09973abe5dadfc5ba FMC...	Off	10.0.101.0/24	-	-	251	ca-central-1b

What it Shows:

This screenshot is from the **AWS VPC Dashboard**, specifically under **"Subnets."**

How This Was Done:

1. VPC Created:

- From AWS Console: Go to **VPC > Create VPC**
- CIDR Block: **10.0.0.0/16**
- Named it **FMCG-VPC**

2. Subnets Created, chose "Create Subnet"

- Assigned CIDR ranges like **10.0.1.0/24**, **10.0.2.0/24**, etc.
- Named each subnet according to the department (HR, IT, etc.)
- Selected AZs (Availability Zones): e.g., **ca-central-1a** or **1b**

3. Tagged for clarity: Each subnet includes naming conventions and roles (Public/Private).

Why This Matters:

Each **department** gets its own subnet for logical isolation. Public subnets (Public-A, Public-B) host resources like NAT. Private subnets ensure internal systems like finance for example, stay isolated from direct internet access.

Cost Estimation Over 10 Years

AWS VPC Design - 10-Year Cost Chart

Component	Description	Quantity / Usage	Rate	Total (10 Years)	Notes
VPC	Base VPC, subnets, route tables	N/A	Free	\$0	No cost for setup or configuration
Internet Gateway	For public subnets to access the internet	1 IGW	Free	\$0	Free when attached to VPC
Elastic IP (EIP)	For NAT Gateway	1 EIP (attached)	Free	\$0	Free when attached to running resource
NAT Gateway (Hourly)	Enables private subnets to access the internet	87,600 hours	\$0.045/hour	\$3,942.00	24/7 usage assumed for 10 years
NAT Data Processing	Data sent from private subnet to internet	12,000 GB (100 GB/month)	\$0.045/GB	\$540.00	Based on estimated usage
Total Cost				\$4,482.00	Over 10 years

Summary of Current Setup

- **VPC Name:** FMCG-VPC
- **CIDR Block:** 10.0.0.0/16
- **Subnets:** 14 total (11 used for departments + 3 extra/unknown), split between:
 - **ca-central-1a, ca-central-1b**

Subnet Sections:

- **Public Subnets:** Public-A, Public-B
- **Private Subnets for Departments:** Sales-Private, Legal-Private, HR-Private, etc.

Route Tables:

- **Public-RT** - for public subnets
- **Private-RT** - for private subnets

Network Connections:

- **NAT Gateway:** FMCG-NAT
- **Internet Gateway:** FMCG-IGW

Breakdown of Each AWS VPC Setup

1. VPC, Subnets, Route Tables

- **Cost: \$0** (Free)
- **Explanation:** Creating a VPC, subnets (both public/private), and route tables does **not incur any hourly or usage charges**. You can create and configure multiple subnets and route tables at no cost.

2. Internet Gateway (IGW)

- **Cost: \$0** (Free)
- **Explanation:** Internet Gateways are **free to attach and use**. You only pay for the data transfer **from EC2 instances** or other services using the IGW, but **not for the IGW itself**.

3. Elastic IP (EIP) for NAT Gateway

- **Cost: \$0** (since it's attached)
- **Explanation:** An Elastic IP **is free as long as it is attached to a running resource**, like your NAT Gateway. AWS **only charges** if the EIP is **allocated but not associated** with a resource or if it is associated but the instance is stopped.

NAT Gateway vs Internet Gateway

Feature	NAT Gateway	Internet Gateway
Subnet Type	Private	Public
Outbound Internet	Yes	Yes
Inbound Internet	No	Yes (if public IP)
Main Purpose	Outbound for private subnets	Full access for public subnets

4. NAT Gateway

Two types of charges:

a. Hourly Charge

This is the **base charge** AWS bills you **just for running the NAT Gateway**, even if **no data is transferred**.

- **Rate:** \$0.045 per hour
- **Time Frame:** 24 hrs × 365 days × 10 years = **87,600 hours**
- **Cost:** \$0.045 × 87,600 = \$3,942.00

b. Data Processing Cost

This is the cost for **data passing through the NAT Gateway** (from private subnets to the internet). You only pay for **outbound traffic** – the amount of data **processed and forwarded**

- **Estimation:** 100 GB/month
- **Total Over 10 Years:** 100 GB × 12 × 10 = **12,000 GB**
- **Rate:** \$0.045 per GB
- **Cost:** \$0.045 × 12,000 GB = \$540.00

Comparative Analysis

In evaluating the most suitable network infrastructure for the organization’s evolving operational needs, two primary approaches were assessed: a traditional on-premises Cisco-based infrastructure, and a fully cloud-native solution using Amazon Web Services (AWS) Virtual Private Cloud (VPC). This comparative analysis considers a 10-year cost projection—including hardware end-of-sale (EOS) and end-of-life (EOL) timelines, internet service provider (ISP) dependencies, and ongoing support costs—as well as critical factors such as security, scalability, performance, and maintenance.

The goal is to identify the optimal solution that balances current business requirements with long-term sustainability and growth.

Cost Analysis

On-Premises (Cisco) Solution

The on-premises Cisco network requires a **significant upfront investment in hardware**, including switches, routers, and firewalls, totaling \$135,364.75**. Additionally, **setup labor costs** \$4,800**, and **10-year maintenance and replacement** costs bring the total expenditure to \$343,211.90**. While this model has predictable long-term costs, it requires capital expenditure (CapEx), which may strain budgets for mid-sized companies.

AWS Cloud Solution

The AWS VPC model has **minimal upfront costs**, with expenses primarily coming from NAT Gateway usage (\$3,942.00** over 10 years) and data processing (\$540.00**). However, **operational expenditures (OpEx) could increase if the company scales rapidly**, requiring additional AWS services (e.g., EC2 instances, S3 storage). Unlike on-premises, AWS eliminates hardware replacement costs, but long-term expenses depend on usage growth.

Cost Comparison Table

Factor	On-Premises (Cisco)	AWS Cloud
Upfront Cost	\$135,364.75 (hardware)	\$0 (no hardware purchase)
Ongoing Costs	\$67,682.40 (maintenance)	\$4,482 (NAT + data transfer)
Scalability Cost	Additional hardware purchases	Pay-as-you-go model
10-Year Total Cost	\$343,211.90	\$4,482 (minimal baseline)

Key Insight:

- **On-premises is cost-effective** if the company has **stable growth** and prefers **fixed expenses**.
- **AWS is cheaper initially** but may become expensive with **high data transfer or compute needs**.

Scalability & Flexibility

On-Premises Limitations

Scaling an on-premises network requires purchasing additional switches, routers, and licenses, which can take weeks to deploy. If the Customer Service department grows by 20%, the company must pre-emptively buy hardware, leading to over-provisioning or bottlenecks.

AWS Elasticity

AWS allows instant scaling—new subnets, virtual servers, and storage can be provisioned in minutes. If the company expands, AWS automatically adjusts resources without upfront purchases. Additionally, multi-region deployment is possible if the company opens new offices.

Scalability Comparison Table

Factor	On-Premises (Cisco)	AWS Cloud
Expansion Speed	Weeks (hardware procurement)	Minutes (cloud provisioning)
Handling Peak Demand	Requires over-provisioning	Auto-scaling available
Global Expansion	Complex (requires new sites)	Seamless (multi-region VPC)

Key Insight:

- **AWS is superior for rapid growth** but may require **cloud networking expertise**.
- **On-premises is rigid** but offers **predictable performance** for stable workloads.

Security & Compliance

On-Premises Security

The company has full physical control over firewalls, switches, and data. VLAN segmentation ensures departments (e.g., Legal, Finance) remain isolated. However, security depends on internal IT expertise, and breaches could go undetected without proper monitoring.

AWS Shared Responsibility Model

AWS provides built-in DDoS protection, IAM policies, and encryption, but the company must configure security groups, NACLs, and logging. Misconfigurations (e.g., open S3 buckets) are a common cloud risk.

Security Comparison Table

Factor	On-Premises (Cisco)	AWS Cloud
Physical Security	Full control (on-site)	AWS-managed data centers
Network Segmentation	VLANs + ACLs	Security Groups + NACLs
Compliance	Easier for strict regulations	Requires AWS compliance config

Key Insight:

- **On-premises is better for highly regulated data** (e.g., financial records).
- **AWS is secure but requires proper setup** to avoid cloud vulnerabilities.

Performance & Reliability

On-Premises LAN Performance

- 10 Gb Ethernet ensures low-latency internal traffic (critical for large file transfers).
- No internet dependency—ideal for real-time applications (VoIP, video conferencing).

AWS Cloud Performance

- Internet latency may affect internal communications.
- AWS backbone is fast, but ISP reliability impacts performance.

Reliability Comparison Table

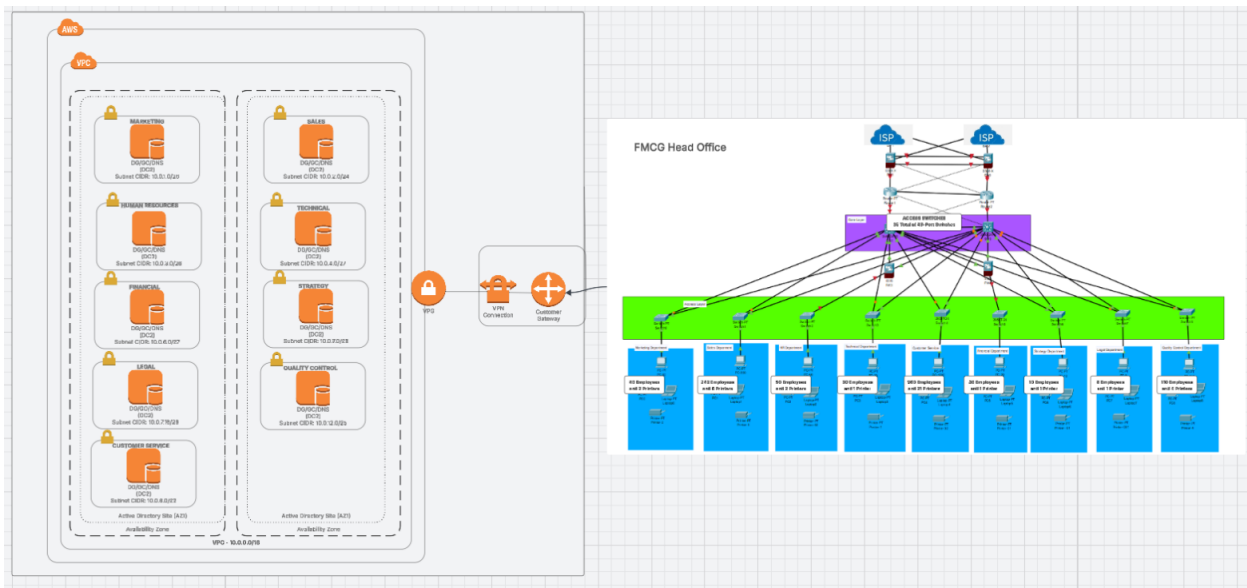
Factor	On-Premises (Cisco)	AWS Cloud
Uptime	Depends on redundancy setup	99.99% SLA (multi-AZ)
Disaster Recovery	Manual backups needed	Automated snapshots + S3
Single Point of Failure	Yes (if no redundancy)	No (AWS high availability)

Key Insight:

- **AWS offers better uptime guarantees, but on-premises provides faster LAN speeds.**

Final Recommendation: Hybrid Network Solution

After a thorough analysis of both on-premises (Cisco) and AWS cloud solutions, our team recommends a **hybrid network architecture** as the most strategic approach for the FMCG company. This model combines the best of both worlds, leveraging the performance and control of on-premises infrastructure while integrating the scalability and flexibility of AWS cloud services. Below, we provide an in-depth justification for this recommendation.



Justification for the Recommended Approach

1. Business Needs Alignment

The FMCG company has 1,258 employees with expected growth, particularly in Customer Service (20% expansion). A purely on-premises solution would require proactive hardware investments to accommodate growth, leading to over-provisioning or bottlenecks. Conversely, a full cloud migration could introduce latency in internal workflows (e.g., large file transfers between departments).

Why Hybrid?

- **Core departments (Sales, Customer Service, Quality Control)** remain **on-premises** for **low-latency, high-bandwidth operations**.
- **AWS cloud** is used for **scalable workloads** (e.g., seasonal demand spikes, remote workforce support).

2. Cost Efficiency Over 10 Years

On-Premises Cost Challenges

- \$335K total cost over 10 years (hardware + maintenance).
- Replacement cycles (Years 5, 6, 9) introduce unplanned CapEx spikes.

AWS Cost Benefits

- \$4.5K baseline cost (NAT Gateway + data transfer).
- Pay-as-you-go scaling avoids upfront hardware purchases.

Hybrid Cost Advantage:

- **Reduces long-term CapEx** by shifting variable workloads (backups, remote) to AWS.
- **Avoids over-investing in on-premises hardware** for uncertain growth.

3. Security & Compliance Balance

On-Premises Security Strengths

- Full physical control over Finance, Legal, and HR data.
- No third-party dependency for compliance (e.g., data sovereignty).

AWS Security Enhancements

- Built-in encryption, IAM, and DDoS protection.
- Automated security patches reduce IT workload.

Hybrid Security Advantage:

- **Critical data stays on-premises** (avoiding cloud misconfiguration risks).
- **Non-sensitive workloads (marketing, external apps) use AWS** with enterprise-grade security.

4. Performance Optimization

Workload Type	Recommended Solution	Reason
Internal file sharing	On-premises 10 Gb LAN	Ultra-low latency
Remote work VPN	AWS Client VPN	Secure external access
Disaster recovery	AWS S3 + Glacier	Automated, geo-redundant

Why This Mix?

- **On-premises ensures real-time performance** for daily operations.
- **AWS handles burst capacity** without requiring permanent hardware.

5. Future-Proofing the Network

The hybrid model future-proofs the company by:

1. Allowing gradual cloud adoption (no "all-or-nothing" migration risk).
2. Supporting multi-office expansion via AWS VPC peering.
3. Enabling AI/ML integrations (e.g., AWS SageMaker for sales analytics).

Risk Mitigation:

- If cloud costs escalate, the company **can rebalance workloads back on-premises**.
- If on-premises hardware ages, critical systems **can failover to AWS**.

Implementation Roadmap

Phase 1 (Year 0-1): On-Premises Core

- Deploy **Cisco switches/routers** for primary LAN.
- Migrate **print servers, internal databases** on-prem.

Phase 2 (Year 2-3): AWS Hybrid Integration

- Implement **AWS Storage Gateway** for backups.
- Deploy **AWS Direct Connect** for secure cloud links.

Phase 3 (Year 5+): Scalability Focus

- Shift **dev/test environments** to AWS.
- Use **AWS Workspaces** for remote teams.

Conclusion

The **hybrid network solution** provides the FMCG company with:

- **Cost control** (reduced CapEx, OpEx flexibility)
- **Performance assurance** (LAN speed + cloud elasticity)
- **Strategic agility** (adaptable to future tech trends)

Final Decision:

- Start with on-premises Cisco for core infrastructure.
- Progressively integrate AWS for scalability and innovation.

This approach **minimizes risk while maximizing long-term value**, aligning perfectly with the company's growth trajectory. It's not just a technical solution—**it's a strategic one**.

References

- Cisco Press. (n.d.). Hierarchical network design overview. Retrieved from <https://www.ciscopress.com/articles/article.asp?p=2202410&seqNum=4>
- CloudMyLab. (n.d.). Star network topology explained: Advantages and use cases. Retrieved from <https://blog.cloudmylab.com/star-network-topology>
- NetworkLessons.com.. (n.d.). Network topologies. Retrieved from <https://networklessons.com/cisco/ccna-routing-switching-icnd1-100-105/network-topologies>
- NetworkInterview.com.. (n.d.). Partial mesh network topology. Retrieved from <https://networkinterview.com/partial-mesh-network-topology/>
- NetworkSanta. (n.d.). Three-tier network architecture: A comprehensive guide. Retrieved from <https://networksanta.com/three-tier-network-architecture-guide/>
- Ques10. (n.d.). Explain FC SAN topologies & FC protocol stack. Retrieved from <https://www.ques10.com/p/20597/explain-fc-san-topologies-fc-protocol-stack-1/>
- StellaStra. (n.d.). Advantages and benefits of a star network topology. Retrieved from <https://stellastr.com/advantages-benefits-of-star-network-topology/>
- Cloudflare. (n.d.). What is subnetting?. Retrieved from <https://www.cloudflare.com/learning/network-layer/what-is-subnetting/>
- Cisco Systems. (n.d.). Cisco Catalyst 8300 Series Edge Platforms: Features and specifications. Retrieved from <https://www.cisco.com/c/en/us/products/routers/catalyst-8300-series-edge-platforms/index.html>
- Cisco Systems. (n.d.). Catalyst 9300 Series Switches: Features and specifications. Retrieved from <https://www.cisco.com/c/en/us/products/switches/catalyst-9300-series-switches/index.html>
- Cisco Systems. (n.d.). Catalyst 9200 Series Switches: Features and specifications. Retrieved from <https://www.cisco.com/c/en/us/products/switches/catalyst-9200-series-switches/index.html>
- Cisco Systems. (n.d.). Cisco Firepower 1000 Series: Features and specifications. Retrieved from <https://www.cisco.com/c/en/us/products/security/firewalls/firepower-1000-series/index.html>
- CDW. (n.d.). Cisco Firepower 1120 Next-Generation Firewall. Retrieved from <https://www.cdw.com/product/cisco-firepower-1120-next-generation-firewall-firewall/5617296>
- Amazon Web Services. (n.d.). Amazon VPC pricing. Retrieved from <https://aws.amazon.com/vpc/pricing/>
- Amazon Web Services. (n.d.). Pricing for NAT gateways. Retrieved from <https://docs.aws.amazon.com/vpc/latest/userguide/nat-gateway-pricing.html>

Amazon Web Services. (n.d.). Architecture with an internet gateway and a NAT gateway. Retrieved from <https://docs.aws.amazon.com/network-firewall/latest/developerguide/arch-igw-ngw.html>

Amazon Web Services. (n.d.). Elastic IP addresses. Retrieved from <https://docs.aws.amazon.com/AWSEC2/latest/UserGuide/elastic-ip-addresses-eip.html>

Amazon Web Services. (n.d.). What is Amazon VPC?. Retrieved from <https://docs.aws.amazon.com/vpc/latest/userguide/what-is-amazon-vpc.html>

Amazon Web Services. (n.d.). What is hybrid cloud architecture?. Retrieved from <https://aws.amazon.com/hybrid/>

Cisco Systems. (n.d.). Hybrid cloud solutions with Cisco and AWS. Retrieved from <https://www.cisco.com/c/en/us/solutions/cloud/hybrid-cloud.html>

SinanSoft. (n.d.). Hybrid cloud: Scalability, security, and cost efficiency. Retrieved from <https://sinansoft.com/blog/hybrid-cloud-is-a-modern-approach-to-scalability-and-flexibility-in-business>

Scale Computing. (n.d.). Hybrid cloud architecture explained. Retrieved from <https://www.scalecomputing.com/resources/hybrid-cloud-architecture-explained>

The Knowledge Academy. (n.d.). What is hybrid cloud strategy? Benefits and challenges. Retrieved from <https://www.theknowledgeacademy.com/blog/hybrid-cloud-strategy/>